

GURRAM KARTHIKEYA

AI Engineer | Python Developer | Full-Stack Developer
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GitHub | LinkedIn | Portfolio

CAREER SUMMARY

Innovative B.Tech CSE (AI & ML) fresher with proven experience designing AI-powered SaaS applications, scalable REST APIs, and LLM integration workflows using the Claude API and OpenAI API. Proficient in Python backend engineering with FastAPI, MERN stack development, asynchronous processing, and cloud deployment on Render and Vercel. Demonstrated ability to architect production-ready, multi-service systems — JWT-secured Express backends, MongoDB Atlas persistence layers, React frontends, and Python PDF microservices — and deliver them through structured deployment pipelines. Passionate about solving real-world problems at the intersection of AI engineering and full-stack development.

EDUCATION

B.Tech — CSE (AI & Machine Learning) — St. Mary's Engineering College, Hyderabad	2022 – 2026
Intermediate (MPC) — Sri Chaitanya Junior College <i>Percentage: 63%</i>	2020 – 2022
SSC — Gouthama High School <i>Percentage: 98%</i>	2020

TECHNICAL SKILLS

AI / ML: Supervised & Unsupervised Learning, Scikit-learn, NumPy, Pandas, Matplotlib, Feature Engineering, Model Evaluation, NLP Basics, Prompt Engineering

Programming: Python 3, JavaScript (ES6+), Java, C, C++ — OOP, Exception Handling, Data Analysis

AI Integrations: Claude API (Anthropic), OpenAI API, LLM Workflows, RAG Pipelines, Agentic Workflows, Structured Output Generation, NLP Pipelines

Backend & APIs: FastAPI, Node.js, Express.js, REST API Design, JWT Authentication, Async Endpoints, WeasyPrint PDF Microservice

Frontend: React.js (v18), HTML5, CSS3, Tailwind CSS, Vite, React Router, Responsive UI Design

Databases: MongoDB Atlas, SQL, SQLite, Mongoose ORM, DBMS Concepts

Tools & Cloud: Git, GitHub, Postman, VS Code, Linux/Termux, Render, Vercel, Netlify, GitHub Codespaces, Docker (basics)

CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS

PROJECTS

InvoiceAI — AI Invoice & Document Generator SaaS **GitHub** ■

React.js · Node.js · Express.js · FastAPI · MongoDB Atlas · Claude API · JWT · WeasyPrint · Render · Vercel

- Built a full-stack AI-powered SaaS platform that automates invoice, quotation, receipt, and purchase-order workflows — users describe a document in plain English and receive a professionally formatted PDF in seconds via the **Claude API**.
- Architected a scalable **three-tier system**: React 18 + Vite frontend, Node.js/Express REST backend, and a Python FastAPI microservice with WeasyPrint for high-fidelity PDF generation.
- Implemented secure **JWT authentication**, MongoDB Atlas persistence with Mongoose schemas, and a business profile system that auto-populates all AI-generated documents.
- Developed a dynamic line-item editor with automated **GST and totals calculations**, an analytics revenue dashboard, and advanced document filtering with full CRUD management.
- Configured cloud deployment pipelines — backend and PDF service on **Render**, React/Vite frontend on **Vercel** — with optimised async API throughput for production-ready performance.

Box Cricket Booking System Live ■ | Frontend ■ | Backend ■

React.js · Node.js · Express.js · MongoDB · Render

- Developed a production-ready real-time slot-booking platform with availability tracking, conflict detection logic, and automated booking validation for a live sports academy.
- Designed and tested **10+ RESTful API endpoints** in Postman; integrated Twilio SMS booking confirmations and a Recharts admin analytics dashboard.
- Built a mobile-responsive React frontend with seamless backend integration; deployed the full-stack application on Render.

Web Scraping & Data Analysis Tool GitHub ■

Python · BeautifulSoup · Pandas · Matplotlib

- Automated structured data extraction from multiple web sources using BeautifulSoup; built reusable scraping pipelines with retry logic, rate limiting, and robust error handling.
- Performed data preprocessing, transformation, and exploratory analysis with Pandas; produced analytical visualisation reports with Matplotlib across a complete ML-style data pipeline.

AI / ML KNOWLEDGE AREAS

- **Supervised Learning:** Linear Regression, Logistic Regression, Decision Trees, KNN — Accuracy, Precision, Recall, F1-Score Evaluation
- **Unsupervised Learning:** K-Means Clustering, PCA, Dimensionality Reduction
- **Data & NLP Pipelines:** Feature Engineering, Data Preprocessing, Tokenisation, NLP Preprocessing, Model Training Pipelines
- **AI Integration & Deployment:** Claude API, OpenAI API, Prompt Engineering, FastAPI ML Deployment, REST API Wrapping for ML Models

CERTIFICATIONS

- Data Analytics Job Simulation — **Deloitte (Forage)** (2025)
- Solutions Architecture Job Simulation — **AWS (Forage)** (2025)
- Full Stack with AI Bootcamp — **NoviTech R&D Pvt. Ltd.** (2024)